Lecture 14

Representative functions and tensor reconstruction

Objectives. *This lecture asks what information about a compact Lie group K is contained in all its representations. We collect all matrix coefficients into one algebra $\mathscr{R}(K)$. The group law becomes three maps on this algebra. We then state how K is recovered, and what is recovered when the unitary structure is forgotten.*

By the end of this lecture you will be able to:

- Prove that $\mathscr{R}(K)$ is a $*$-algebra whose products come from tensor products and whose conjugation comes from conjugate representations.
- Define Δ, ε, S and prove the Hopf algebra identities, also in matrix form.
- State Tannaka–Krein reconstruction with the unitary fibre functor, and contrast it with the complex reconstruction of $K_{\mathbb{C}}$.
- Compute these structures for $\mathrm{U}(1)$ and $\mathrm{SU}(2)$, and explain why $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category when K is infinite.

A space can be replaced by the algebra of functions on it. A group has more: its product appears as a map on these functions. This description still makes sense for algebras that are not commutative, and Lecture 28 returns to them. Nothing here depends on that.

Throughout, K is a compact Lie group. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ has as objects the finite-dimensional continuous complex representations (Π_V, V). Its morphisms are the K-equivariant linear maps.

14.1 Matrix coefficients

Definition 14.1. Let V be an object of $\mathrm{Rep}_{\mathrm{fd}}(K)$, $\lambda \in V^*$ and $v \in V$. The *matrix coefficient* $c^V_{\lambda,v} \colon K \to \mathbb{C}$ is

$$c^V_{\lambda,v}(g) = \lambda\big(\Pi_V(g)v\big).$$

The space $\mathscr{R}(K)$ of *representative functions* is the linear span of all matrix coefficients of all objects V.

This is the coefficient space of Lecture 10, since every finite-dimensional continuous representation is unitarisable (Lecture 5).

Fix a basis (e_j) of V with dual basis (e^i). The matrix of $\Pi_V(g)$ in this basis is $D^V(g)$, with entries $D^V_{ij}(g) = e^i(\Pi_V(g)e_j)$. Expanding $\lambda = \sum_i \lambda(e_i)e^i$ and $v = \sum_j e^j(v)e_j$ gives

$$c^V_{\lambda,v} = \sum_{i,j} \lambda(e_i)\, e^j(v)\, D^V_{ij}.$$

So every coefficient of V is a linear combination of the D^V_{ij}. For a K-invariant inner product on V (Lecture 5) we may take $\lambda = \langle u, \cdot\rangle$, which gives $c^V_{\lambda,v}(g) = \langle u, \Pi_V(g)v\rangle$.

The *conjugate representation* $\overline{V}$ is the additive group V with scalar multiplication $t \cdot v = \bar{t}v$ and the same operators $\Pi_V(g)$. These operators are linear on $\overline{V}$:

$$\Pi_V(g)(t \cdot v) = \Pi_V(g)(\bar{t}v) = \bar{t}\,\Pi_V(g)v = t \cdot \Pi_V(g)v.$$

For $\lambda \in V^*$ put $\bar{\lambda}(v) = \overline{\lambda(v)}$. The functional $\bar{\lambda}$ is linear on $\overline{V}$:

$$\bar{\lambda}(t \cdot v) = \overline{\bar{t}\,\lambda(v)} = t\,\bar{\lambda}(v).$$

A basis (e_j) of V is also a basis of $\overline{V}$, with dual basis $(\overline{e^i})$. Hence

$$D^{\overline{V}}_{ij}(g) = \overline{e^i}\big(\Pi_V(g)e_j\big) = \overline{D^V_{ij}(g)}, \qquad \text{so} \qquad D^{\overline{V}}(g) = \overline{D^V(g)}.$$

Proposition 14.2. *Let $\lambda \in V^*$, $v \in V$, $\mu \in W^*$, $w \in W$ and $h \in K$. Then:*

(a) $c^V_{\lambda,v} + c^W_{\mu,w} = c^{V\oplus W}_{(\lambda,\mu),(v,w)}$;

(b) $c^V_{\lambda,v}\, c^W_{\mu,w} = c^{V\otimes W}_{\lambda\otimes\mu,\, v\otimes w}$;

(c) $\overline{c^V_{\lambda,v}} = c^{\overline{V}}_{\bar{\lambda},v}$;

(d) $g \mapsto c^V_{\lambda,v}(gh)$ *equals* $c^V_{\lambda,\Pi_V(h)v}$, *and* $g \mapsto c^V_{\lambda,v}(hg)$ *equals* $c^V_{\lambda\circ\Pi_V(h),v}$.

Hence $\mathscr{R}(K)$ is the set of matrix coefficients. It is a subalgebra of $C(K)$ containing the constants, closed under conjugation and translations.

Proof. Matrix coefficients are continuous because $g \mapsto \Pi_V(g)$ is continuous. The sum rule (a) holds because $\Pi_{V\oplus W}(g)$ acts componentwise. For the product rule (b), the tensor product representation gives

$$(\lambda \otimes \mu)\big(\Pi_V(g)v \otimes \Pi_W(g)w\big) = \lambda(\Pi_V(g)v)\,\mu(\Pi_W(g)w).$$

The conjugation rule (c) is the definition of $\bar{\lambda}$. The translation rule (d) uses $\Pi_V(gh) = \Pi_V(g)\Pi_V(h)$ and $\Pi_V(hg) = \Pi_V(h)\Pi_V(g)$.

Scalar multiples are coefficients, since $t\,c^V_{\lambda,v} = c^V_{t\lambda,v}$. By the sum rule (a), sums of coefficients are coefficients. So the span consists of coefficients. By the product, conjugation and translation rules (b)–(d), this set is closed under products, conjugation and translations. The constant 1 is $c^{\mathbb{C}}_{1,1}$ for the trivial representation. ∎

We use the Peter–Weyl theorem for compact Lie groups from Lecture 10. Choose one representative V for each class in $\mathrm{Irr}(K)$ and a basis of each. Then

$$\mathscr{R}(K) = \bigoplus_{[V]\in\mathrm{Irr}(K)} \operatorname{span}\{D^V_{ij}\},$$

an algebraic direct sum whose summand for V has dimension $(\dim V)^2$. Moreover $\mathscr{R}(K)$ is dense in $C(K)$ for the supremum norm and separates points.

14.2 The group law as structure maps

The group law is a map $K \times K \to K$. Pulling functions back along the group law produces functions of two variables. We first show that a tensor of functions is determined by the function of two variables that it defines.

Lemma 14.3. *Let A and B be linear spaces of functions on sets X and Y. The linear map from $A \otimes B$ to functions on $X \times Y$ with $f \otimes h \mapsto \big((x,y) \mapsto f(x)h(y)\big)$ is injective.*

Proof. Every tensor can be written $F = \sum_i f_i \otimes h_i$ with $h_1, \dots, h_n$ linearly independent. Suppose $\sum_i f_i(x)h_i(y) = 0$ for all x, y. Fix x. Then $\sum_i f_i(x)h_i$ is the zero function on Y. The numbers $f_i(x)$ are the coefficients of a vanishing combination of independent functions, so $f_i(x) = 0$ for all i. As x was arbitrary, each $f_i = 0$ and $F = 0$. ∎

Applying the lemma twice handles three factors. So we identify $\mathscr{R}(K)^{\otimes 2}$ and $\mathscr{R}(K)^{\otimes 3}$ with spaces of functions on K^2 and K^3. The identification is an algebra homomorphism for $(f \otimes h)(f' \otimes h') = ff' \otimes hh'$. Indeed both sides give the function $(x,y) \mapsto f(x)f'(x)h(y)h'(y)$.

Definition 14.4. A *Hopf algebra* is a unital algebra A with unital algebra homomorphisms $\Delta \colon A \to A \otimes A$, $\varepsilon \colon A \to \mathbb{C}$ and a linear map $S \colon A \to A$ such that

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta, \qquad (\varepsilon \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon)\Delta,$$
$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta,$$

where m is multiplication and $\eta(t) = t1$. A *Hopf $*$-algebra* is a Hopf algebra with an antilinear map $*$ such that $f^{**} = f$, $(fh)^* = h^* f^*$, and Δ, ε preserve $*$. Here $(f \otimes h)^* = f^* \otimes h^*$ and $t^* = \bar{t}$ on $\mathbb{C}$.

For $f \in \mathscr{R}(K)$ and $g, h \in K$ define

$$\Delta f(g,h) = f(gh), \qquad \varepsilon f = f(e), \qquad Sf(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

For the coordinate function z on $\mathrm{U}(1)$ we have $\Delta z^n = z^n \otimes z^n$, since $(gh)^n = g^n h^n$. For a matrix entry we will find $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$, which is matrix multiplication. In general, a priori Δf is only a function on $K \times K$. The first task is to show that it is a tensor in $\mathscr{R}(K)^{\otimes 2}$.

Theorem 14.5. *These maps make $\mathscr{R}(K)$ a commutative Hopf $*$-algebra, and $S(f^*) = (Sf)^*$. For every V and basis (e_j),*

$$\Delta D^V_{ij} = \sum_k D^V_{ik} \otimes D^V_{kj}, \qquad \varepsilon D^V_{ij} = \delta_{ij}, \qquad SD^V_{ij}(g) = \big(D^V(g)^{-1}\big)_{ij}. \tag{14.1}$$

For an orthonormal basis of a K-invariant inner product, $SD^V_{ij} = (D^V_{ji})^*$.

Proof. Codomains. Insert the expansion $v = \sum_k e^k(v)e_k$ for the vector $v = \Pi_V(h)e_j$. This gives

$$D^V_{ij}(gh) = e^i\big(\Pi_V(g)\Pi_V(h)e_j\big) = \sum_k e^i(\Pi_V(g)e_k)\, e^k(\Pi_V(h)e_j).$$

So ΔD^V_{ij} is the function of the tensor $\sum_k D^V_{ik} \otimes D^V_{kj}$. This tensor is unique by the tensor injectivity lemma, Lemma 14.3. The same insertion for a general coefficient gives

$$\Delta c^V_{\lambda,v} = \sum_k c^V_{\lambda,e_k} \otimes c^V_{e^k,v}. \tag{14.2}$$

Hence Δ maps $\mathscr{R}(K)$ into $\mathscr{R}(K)^{\otimes 2}$. Let $\Pi_{V^*}(g)\lambda = \lambda \circ \Pi_V(g)^{-1}$ be the dual representation. Let $\hat{e}_j \in V^{**}$ be evaluation at e_j. Then

$$SD^V_{ij}(g) = e^i\big(\Pi_V(g)^{-1}e_j\big) = \hat{e}_j\big(\Pi_{V^*}(g)e^i\big) = c^{V^*}_{\hat{e}_j,e^i}(g).$$

So S preserves $\mathscr{R}(K)$. Also $\varepsilon D^V_{ij} = e^i(e_j) = \delta_{ij}$, and $f^* \in \mathscr{R}(K)$ by the conjugation rule in Proposition 14.2.

Algebra maps. As functions, $\Delta(ff')(g,h) = f(gh)f'(gh) = (\Delta f\,\Delta f')(g,h)$. The identification is an injective algebra homomorphism, so $\Delta(ff') = \Delta f\,\Delta f'$. Also $\Delta 1 = 1 \otimes 1$. Pointwise products show that ε and S are unital algebra homomorphisms.

Identities. Write $\Delta f = \sum_i f_i \otimes f'_i$, so $\sum_i f_i(g)f'_i(h) = f(gh)$ for $g,h \in K$. As functions of $(g,h,k') \in K^3$,

$$(\Delta \otimes \mathrm{id})\Delta f\,(g,h,k') = \textstyle\sum_i f_i(gh)f'_i(k') = f\big((gh)k'\big),$$
$$(\mathrm{id} \otimes \Delta)\Delta f\,(g,h,k') = \textstyle\sum_i f_i(g)f'_i(hk') = f\big(g(hk')\big).$$

These agree by associativity, so the tensors agree by the tensor injectivity lemma, Lemma 14.3. For the counit, $(\varepsilon \otimes \mathrm{id})\Delta f = \sum_i f_i(e)f'_i$ is the function $h \mapsto f(eh) = f(h)$. For the antipode,

$$m(S \otimes \mathrm{id})\Delta f\,(g) = \sum_i f_i(g^{-1})f'_i(g) = f(g^{-1}g) = f(e).$$

The mirror identities use $he = h$ and $gg^{-1} = e$. Each group axiom is used once.

$*$-structure. Conjugation is antilinear and involutive, and $\mathscr{R}(K)$ is commutative. The function $\Delta(f^*)(g,h) = \overline{f(gh)}$ is that of $(\Delta f)^*$. Also $\varepsilon(f^*) = \overline{f(e)}$ and $S(f^*)(g) = \overline{f(g^{-1})}$. Finally, if $D^V(g)$ is unitary then $(D^V(g)^{-1})_{ij} = \overline{D^V_{ji}(g)}$. ∎

On matrix coefficients the axioms become matrix identities. We write $D = D^V$ and evaluate each side at group elements:

$$\begin{aligned}
\text{coassociativity:} &\quad \textstyle\sum_{k,l} D_{ik}(g)D_{kl}(h)D_{lj}(k') = \big(D(g)D(h)D(k')\big)_{ij},\\
\text{counit:} &\quad \textstyle\sum_k \delta_{ik}D_{kj}(g) = \big(D(e)D(g)\big)_{ij} = D_{ij}(g),\\
\text{antipode:} &\quad \textstyle\sum_k (SD_{ik})(g)\,D_{kj}(g) = \big(D(g)^{-1}D(g)\big)_{ij} = \delta_{ij}.
\end{aligned}$$

So the three identities encode $D(ghk') = D(g)D(h)D(k')$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. In an orthonormal basis the antipode identity reads

$$\sum_k (D_{ki})^* D_{kj} = \delta_{ij}. \tag{14.3}$$

In other words, the columns of $D(g)$ are orthonormal. Hence S and $*$ together encode the unitarity of $D(g)$. Exercise 14.7 checks these identities on the spin 1 representation of SU(2).

14.3 Recovering K

Definition 14.6. A *character* of $\mathscr{R}(K)$ is a unital algebra homomorphism $\psi\colon \mathscr{R}(K) \to \mathbb{C}$. It is a *$*$-character* if $\psi(f^*) = \overline{\psi(f)}$ for all f.

This algebraic use of the word differs from the trace character of a representation and from the characters $T \to \mathrm{U}(1)$ of a torus T (Lecture 7).

For $g \in K$ the evaluation $\mathrm{ev}_g(f) = f(g)$ is a $*$-character.

Proposition 14.7. *The characters form a group $\mathcal{X}(K)$ with*

$$\psi_1\psi_2 = (\psi_1 \otimes \psi_2) \circ \Delta, \qquad 1 = \varepsilon, \qquad \psi^{-1} = \psi \circ S.$$

The $*$-characters form a subgroup $\mathcal{X}_*(K)$, and $g \mapsto \mathrm{ev}_g$ is an injective homomorphism $K \to \mathcal{X}_*(K)$.

Proof. Since Δ, $\psi_1 \otimes \psi_2$ and S are algebra homomorphisms, $\psi_1\psi_2$ and $\psi \circ S$ are characters. Applying $\psi_1 \otimes \psi_2 \otimes \psi_3$ to coassociativity gives $(\psi_1\psi_2)\psi_3 = \psi_1(\psi_2\psi_3)$. The counit identity gives $\varepsilon\psi = \psi\varepsilon = \psi$. Multiplicativity means $\psi \circ m = m_{\mathbb{C}} \circ (\psi \otimes \psi)$, so

$$(\psi \circ S)\psi = \psi \circ m \circ (S \otimes \mathrm{id}) \circ \Delta = \psi \circ \eta \circ \varepsilon = \varepsilon.$$

The mirror identity gives $\psi(\psi \circ S) = \varepsilon$. The relations $\Delta(f^*) = (\Delta f)^*$ and $S(f^*) = (Sf)^*$ show that products and inverses of $*$-characters are $*$-characters. Finally $(\mathrm{ev}_g \otimes \mathrm{ev}_h)\Delta f = f(gh)$. Injectivity holds because $\mathscr{R}(K)$ separates points. ∎

In other words, Δ, ε and S give the multiplication, the identity and the inverses in $\mathcal{X}(K)$. To recover K we must know that every $*$-character is an evaluation. We now describe the same question in categorical terms.

An element $g \in K$ gives one matrix $\Pi_V(g)$ on each representation V. These matrices are compatible with every intertwiner and with tensor products, and they are unitary for invariant inner products. We now ask whether every family of unitary matrices with these properties comes from an element of K.

Definition 14.8. The *fibre functor* $\omega\colon \mathrm{Rep}_{\mathrm{fd}}(K) \to \mathrm{Vect}_{\mathbb{C}}$ sends each object to its underlying space and each morphism to itself. A *tensor automorphism* of ω is a family $\theta = (\theta_V)$ with $\theta_V \in \mathrm{GL}(V)$ for each object V, such that

$$\theta_W T = T\theta_V \quad (T\colon V \to W \text{ a morphism}), \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

A tensor automorphism θ is *unitary* if θ_V is unitary for every K-invariant inner product on every V.

On $\mathrm{SU}(2)$, the decomposition $V_1 \otimes V_1 = V_2 \oplus V_0$ expresses θ_{V_2} through θ_{V_1}. So θ_{V_1} determines the family, as Example 14.12 shows.

Tensor automorphisms form a group $\mathrm{Aut}_{\otimes}(\omega)$ under $(\theta\theta')_V = \theta_V\theta'_V$, with identity (id_V) and inverse (θ_V^{-1}). Indeed, products of natural families are natural, and naturality $\theta_W T = T\theta_V$ gives $T\theta_V^{-1} = \theta_W^{-1}T$. Products and inverses also preserve the tensor and unit conditions.

Products and inverses of unitary operators are unitary, so the unitary tensor automorphisms form a subgroup $\mathrm{Aut}^{u}_{\otimes}(\omega)$.

The data of the unitary fibre functor are the following.

(i) The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ with its tensor product and unit object $\mathbb{C}$.
(ii) The functor ω, which remembers the underlying vector spaces and linear maps.
(iii) On each $\omega(V)$, the set of K-invariant inner products. On the algebra side these inner products correspond to the $*$-structure.

Each $g \in K$ gives the family with $\theta_V = \Pi_V(g)$. This family is natural because morphisms are equivariant. It is monoidal by the definition of $V \otimes W$. It is unitary because the inner products are K-invariant.

Tensor automorphisms and characters are related by the map $\theta \mapsto \psi_\theta$ given by

$$\psi_\theta\big(c^{V}_{\lambda,v}\big) = \lambda(\theta_V v). \tag{14.4}$$

Exercises 14.4 and 14.5 show that $\theta \mapsto \psi_\theta$ is a group isomorphism $\mathrm{Aut}_{\otimes}(\omega) \to \mathcal{X}(K)$. They also show that θ is unitary exactly when ψ_θ is a $*$-character. In any basis of V, the matrix of θ_V is $\big(\psi_\theta(D^{V}_{ij})\big)_{ij}$.

So a $*$-character ψ gives one operator θ_V on each V, determined by $\psi = \psi_\theta$. The operator θ_V is unitary for every K-invariant inner product on V. Hence its matrix in an orthonormal basis is unitary, while its matrix in a general basis need not be. The hard step of reconstruction is to find a single $g \in K$ whose operators $\Pi_V(g)$ equal the operators θ_V for all V at once. The $\mathrm{SU}(2)$ model, Example 14.12, finds g directly. In general we cite this step.

Theorem 14.9 (Tannaka–Krein duality; cited from BD III.7). *Let K be a compact Lie group. Every $*$-character of $\mathcal{R}(K)$ is ev_g for a unique $g \in K$. Equivalently, $g \mapsto (\Pi_V(g))_V$ is a group isomorphism $K \to \mathrm{Aut}^{u}_{\otimes}(\omega)$.*

The cases to keep in mind for the next theorem are $\mathrm{U}(1) \subset \mathbb{C}^{\times}$ and $\mathrm{SU}(2) \subset \mathrm{SL}(2,\mathbb{C})$, computed in the next section.

Theorem 14.10 (Complexification; cited from BD III.8). *Let K be a compact Lie group and put $K_{\mathbb{C}} = \mathcal{X}(K)$, with the topology of pointwise convergence on $\mathcal{R}(K)$. Then $K_{\mathbb{C}}$ carries a complex manifold structure that makes it a complex Lie group with Lie algebra $\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C}$. The map $g \mapsto \mathrm{ev}_g$ identifies K with a maximal compact subgroup of $K_{\mathbb{C}}$. Every object of $\mathrm{Rep}_{\mathrm{fd}}(K)$ extends uniquely to a holomorphic representation of $K_{\mathbb{C}}$. In algebraic language, $K_{\mathbb{C}}$ is a complex algebraic group with reductive identity component and coordinate ring $\mathcal{R}(K)$. For connected K, it is a connected complex reductive group.*

A map between complex manifolds is *holomorphic* if it is complex differentiable in charts. A *maximal compact subgroup* is a compact subgroup contained in no larger compact subgroup.

Thus plain complex tensor automorphisms recover $\mathrm{Aut}_{\otimes}(\omega) \cong K_{\mathbb{C}}$, not K. The unitary condition, equivalently the $*$-structure, cuts $K_{\mathbb{C}}$ down to K.

14.4 Two models

Example 14.11 (U(1))**.** The irreducible representations of U(1) are the lines $\mathbb{C}_n$, $n \in \mathbb{Z}$, on which z acts by z^n (characters of tori, Lecture 7). The functions z^n are orthonormal for Haar measure (Lecture 5), hence independent. So

$$\mathscr{R}(\mathrm{U}(1)) = \mathbb{C}[z, z^{-1}], \qquad \Delta z = z \otimes z, \quad \varepsilon z = 1, \quad Sz = z^{-1}, \quad z^* = z^{-1}.$$

A character ψ is determined by $w = \psi(z)$. Since $\psi(z)\psi(z^{-1}) = 1$, we have $w \neq 0$. Every $w \in \mathbb{C}^\times$ occurs, via $z^n \mapsto w^n$. The $*$-condition reads $\bar{w} = \psi(z^*) = w^{-1}$, that is $|w| = 1$. Thus $\mathcal{X} = \mathbb{C}^\times$ and $\mathcal{X}_* = \mathrm{U}(1)$.

The categorical computation agrees. Let $\theta \in \mathrm{Aut}_\otimes(\omega)$, and let w_n be the scalar $\theta_{\mathbb{C}_n}$. Every V is the sum of its weight spaces

$$V(n) = \{v \in V : \Pi_V(z)v = z^n v \text{ for all } z \in \mathrm{U}(1)\}.$$

For $v \in V(n)$, the map $T\colon \mathbb{C}_n \to V$, $1 \mapsto v$, is a morphism, so

$$\theta_V v = \theta_V T 1 = T\theta_{\mathbb{C}_n} 1 = w_n v.$$

The isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \to \mathbb{C}_{m+n}$, $1 \otimes 1 \mapsto 1$, gives $w_{m+n} = w_m w_n$, and $w_0 = 1$. Hence θ_V acts by w^n on $V(n)$, with $w = w_1 \in \mathbb{C}^\times$. Weight spaces are orthogonal for every invariant inner product, so θ is unitary exactly when $|w| = 1$. Conversely, each $w \in \mathbb{C}^\times$ occurs, through the character $z^n \mapsto w^n$ and Exercise 14.5. So $\mathrm{Aut}_\otimes(\omega) \cong \mathbb{C}^\times = \mathrm{U}(1)_\mathbb{C}$ and $\mathrm{Aut}^u_\otimes(\omega) \cong \mathrm{U}(1)$.

Example 14.12 (SU(2))**.** Define functions a, b, c, d on SU(2) by

$$g = \begin{pmatrix} a(g) & b(g) \\ c(g) & d(g) \end{pmatrix} = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Here a is the $(1,1)$ entry, as before, but b is now the $(1,2)$ entry. Earlier lectures used b for the $(2,1)$ entry. These functions are the coefficients D_{ij} of $\mathbb{C}^2$ in the standard basis, and

$$ad - bc = 1, \qquad a^* = d, \qquad b^* = -c.$$

Since $\det g = 1$, the inverse matrix is

$$g^{-1} = \begin{pmatrix} d(g) & -b(g) \\ -c(g) & a(g) \end{pmatrix}.$$

So the matrix formulas (14.1) give

$$\begin{aligned} \Delta a &= a \otimes a + b \otimes c, & \Delta b &= a \otimes b + b \otimes d, & Sa &= d, & Sb &= -b, \\ \Delta c &= c \otimes a + d \otimes c, & \Delta d &= c \otimes b + d \otimes d, & Sc &= -c, & Sd &= a. \end{aligned}$$

Also ε sends a, d to 1 and b, c to 0. One checks $SD_{ij} = D^*_{ji}$; for example $Sb = -b = c^*$.

The functions a, b, c, d generate $\mathscr{R}(\mathrm{SU}(2))$ as an algebra. Indeed every irreducible representation is $V_m \cong \mathrm{Sym}^m(\mathbb{C}^2)$ (Lecture 8). In it g sends $e_1^p e_2^{m-p}$ to $(ae_1 + ce_2)^p (be_1 + de_2)^{m-p}$. So each $D_{ij}^{V_m}$ is a polynomial of degree m in a, b, c, d. By the Peter–Weyl theorem these coefficients span $\mathscr{R}(\mathrm{SU}(2))$, which proves the claim.

Let ψ be a $*$-character. Collect its values on the generators in the matrix

$$A = \begin{pmatrix} \psi(a) & \psi(b) \\ \psi(c) & \psi(d) \end{pmatrix}.$$

The relations give

$$\psi(d) = \overline{\psi(a)}, \qquad \psi(c) = -\overline{\psi(b)}, \qquad |\psi(a)|^2 + |\psi(b)|^2 = \psi(ad - bc) = 1.$$

So $A \in \mathrm{SU}(2)$. The algebra maps ψ and ev_A agree on generators, so $\psi = \mathrm{ev}_A$. This verifies Tannaka–Krein duality, Theorem 14.9, for $\mathrm{SU}(2)$ directly.

Without the $*$-condition we only learn $\det A = 1$. Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ gives a character $\psi_A \colon p(a,b,c,d) \mapsto p(A)$ (Exercise 14.3). So $\mathcal{X}(\mathrm{SU}(2)) = \mathrm{SL}(2,\mathbb{C}) = \mathrm{SU}(2)_{\mathbb{C}}$. For $A = \mathrm{diag}(2, \frac{1}{2})$ we get $\psi_A(a) = 2$ but $\psi_A(a^*) = \psi_A(d) = \frac{1}{2} \neq \bar{2}$. The corresponding tensor automorphism has $\theta_{\mathbb{C}^2} = A$, which is not unitary. In physics, this matrix $\mathrm{diag}(2, \frac{1}{2})$ is a boost in the group $\mathrm{SL}(2,\mathbb{C})$, which covers $\mathrm{SO}^+(1,3)$ (Lecture 24).

14.5 The representation category

The Category Theory course studied fusion categories, such as $\mathrm{SU}(2)_k$, with finitely many simple objects. The Clebsch–Gordan rule has the same form. We show that an infinite compact Lie group falls outside this class, and that finiteness is the only axiom that fails.

We use the notions of a rigid and of a symmetric monoidal category from the prerequisite course. Rigidity requires duals, with evaluation and coevaluation maps that satisfy the zigzag identities. A symmetry is a natural isomorphism $\tau_{V,W} \colon V \otimes W \to W \otimes V$ with $\tau_{W,V}\tau_{V,W} = \mathrm{id}$, compatible with associativity.

Definition 14.13. A *fusion category* is a $\mathbb{C}$-linear semisimple rigid monoidal category with finite-dimensional morphism spaces, simple unit object, and finitely many isomorphism classes of simple objects.

The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is semisimple (Lecture 5) with simple unit $\mathbb{C}$ and finite-dimensional morphism spaces. It is rigid, with the dual representation and

$$\mathrm{ev}_V(\lambda \otimes v) = \lambda(v), \qquad \mathrm{coev}_V(1) = \sum_i e_i \otimes e^i.$$

It is symmetric with $\tau(v \otimes w) = w \otimes v$. Exercise 14.6 checks these claims.

Proposition 14.14. *If K is infinite, then $\mathrm{Irr}(K)$ is infinite and $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category. In particular, both conclusions hold when K is connected and nontrivial.*

Proof. We first show that $\mathscr{R}(K)$ is infinite-dimensional. Let $x_1, \dots, x_n \in K$ be distinct. Since $\mathscr{R}(K)$ separates points, for $i \neq j$ we can choose $f_{ij} \in \mathscr{R}(K)$ with $f_{ij}(x_i) \neq f_{ij}(x_j)$. Then

$$h_i = \prod_{j \neq i} \frac{f_{ij} - f_{ij}(x_j)}{f_{ij}(x_i) - f_{ij}(x_j)} \in \mathscr{R}(K).$$

At x_i every factor equals 1. At x_k with $k \neq i$ the factor $j = k$ vanishes. So $h_i(x_k) = \delta_{ik}$. A relation $\sum_i t_i h_i = 0$ evaluated at x_k gives $t_k = 0$. So $h_1, \dots, h_n$ are independent and $\dim \mathscr{R}(K) \geq n$ for every n. If $\mathrm{Irr}(K)$ were finite, Peter–Weyl would give

$$\dim \mathscr{R}(K) = \sum_{[V] \in \mathrm{Irr}(K)} (\dim V)^2 < \infty.$$

So $\mathrm{Irr}(K)$ is infinite. The simple objects of $\mathrm{Rep}_{\mathrm{fd}}(K)$ are the irreducible representations, and a fusion category has only finitely many isomorphism classes of simple objects. So $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category.

Finally, a finite Hausdorff space is discrete. Hence a connected Hausdorff space with at least two points is infinite, and the first part applies to connected nontrivial K. ∎

For a finite group the representation category is a fusion category (Exercise 14.2).

The algebra $\mathscr{R}(K)$ is commutative. Its coproduct is cocommutative only for abelian K. Indeed cocommutativity means

$$f(gh) = f(hg) \quad \text{for all } f \in \mathscr{R}(K),\ g, h \in K.$$

Since $\mathscr{R}(K)$ separates points, this forces $gh = hg$ for all $g, h \in K$.

A later course on quantum groups deforms this classical structure into noncommutative Hopf $*$-algebras. Lecture 28 returns to this comparison. Nothing in this lecture depends on that course.

Self-checks

1. In $\mathscr{R}(\mathrm{U}(1))$ let $f(z) = \mathrm{Re}\, z$. Compute Δf, εf and Sf.
   Answer. $f = \frac{1}{2}(z + z^{-1})$, so $\Delta f = \frac{1}{2}(z \otimes z + z^{-1} \otimes z^{-1})$, $\varepsilon f = 1$ and $Sf = f$. Note $\Delta f \neq f \otimes f$.
2. On $\mathrm{SU}(2)$ write $g \mapsto |g_{11}|^2$ in terms of a, b, c, d. Evaluate ψ_A on it for $A = \mathrm{diag}(2, \frac{1}{2})$.
   Answer. $|\alpha|^2 = aa^* = ad$, and $\psi_A(ad) = 2 \cdot \frac{1}{2} = 1$, while $|A_{11}|^2 = 4$. A non-$*$ character does not evaluate absolute values.
3. For $\mathrm{U}(1)$ let $\theta \in \mathrm{Aut}_{\otimes}(\omega)$ with $\theta_{\mathbb{C}_1} = 3$. Find $\theta_{\mathbb{C}_{-2}}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}}$. Is θ unitary?
   Answer. $\theta_{\mathbb{C}_{-2}} = \frac{1}{9}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}} = \mathrm{diag}(3, \frac{1}{3})$. It is not unitary, since $|3| \neq 1$.

Summary

Matrix coefficients of all finite-dimensional continuous representations of the compact Lie group K form the commutative $*$-algebra $\mathscr{R}(K)$. Tensor products give its products and conjugate representations give its involution. The maps $\Delta f(g, h) = f(gh)$, $\varepsilon f = f(e)$ and $Sf(g) = f(g^{-1})$ make it a Hopf $*$-algebra. On coefficients they are $D(gh) = D(g)D(h)$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. The $*$-characters, or unitary tensor automorphisms of ω, recover K. All characters, or all tensor automorphisms, give $K_{\mathbb{C}}$. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is rigid and symmetric. When K is infinite, it has infinitely many simple objects and is not a fusion category.

Part I in review

Part I asked what a group determines that its Lie algebra does not. A finite-dimensional representation of a Lie algebra integrates to the simply connected group, and it descends to a quotient exactly when the central kernel acts trivially (Lecture 4). This step does not use compactness. For a compact connected group the test reads from the lattice of a maximal torus, and the irreducible representations are labelled by dominant characters (Lecture 7 and Lecture 8). So SU(2) has every V_m, and SO(3) has only those with m even. The lift of a full spatial turn acts on V_m by $(-1)^m$. This sign fixes every ray, so Lecture 13 passes to projective representations.

Compactness gave the rest of Part I. The Haar integral has finite total mass. Averaging gives invariant inner products, complete reducibility and Schur orthogonality (Lecture 5). Averaging a rank-one projection gives a nonzero compact operator, and the Peter–Weyl theorem follows (Lecture 10). So every irreducible unitary representation is finite-dimensional, and $L^2(K)$ is a Hilbert sum of blocks of matrix coefficients. Lecture 11 and Lecture 12 read this as Fourier analysis and as a count of multiplicities on K/H. This lecture recovers K from its coefficient algebra, by a cited theorem.

A noncompact group has no normalised finite Haar integral, so these averaging arguments are no longer available in general. Part II starts from that point.

Reading

BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

BD III.1 for representative functions and their algebra structure. BD III.7 for Tannaka–Krein duality. BD III.8 for the complexification of compact Lie groups. The tensor-category terms are from the prerequisite course.